

NETWORK TRAFFIC ANALYSIS



OBJECTIVES

- Capture Network Traffic.
- Analyze Network packets.
- Identify normal vs Abnormal traffic.
- Detect security threats.
- Apply traffic filtering techniques.
- Improve Network security.

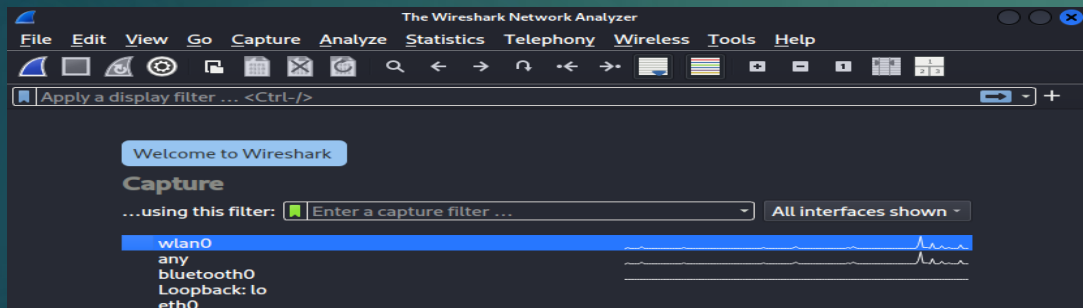


IMPLEMENTATION

Wireshark: Is a network packet analyzer used to capture and analyze data traveling across a network in real time.

Wireshark plays an important role in cybersecurity by allowing analysts to capture and examine network traffic in real time. By inspecting packets from protocols such as TCP, DNS, and HTTP, security professionals can detect suspicious activities like unauthorized connections, malware communication, or network attacks. This helps them investigate incidents, identify vulnerabilities, and protect systems from potential threats.

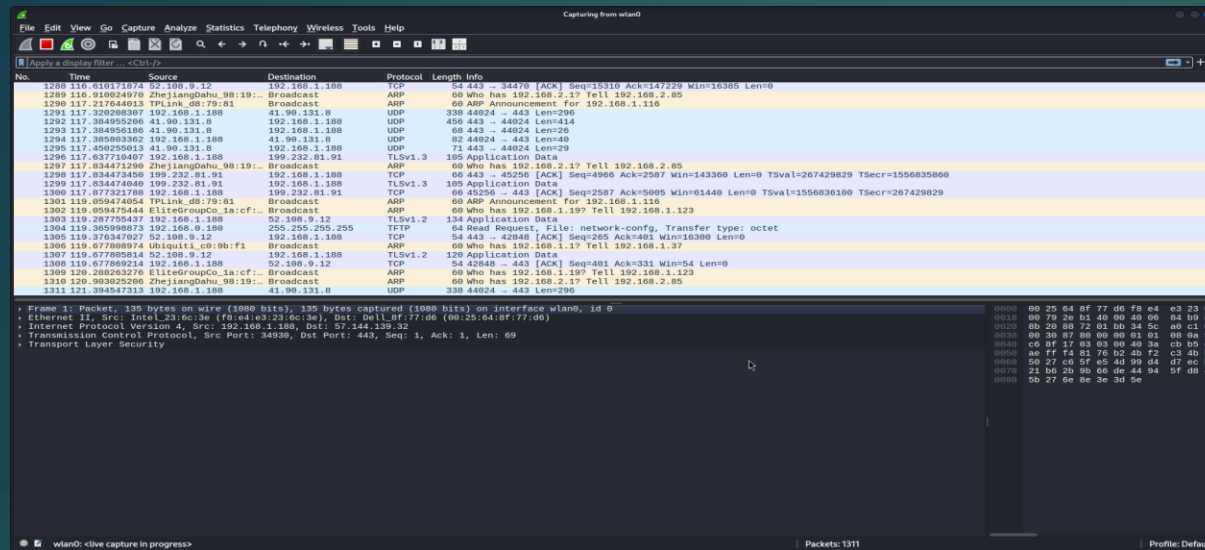
Launching Wireshark, choosing the network interface & starting the capture.



Live capture

4

3/26/2026



We opened a browser and visited the website (testphp.vulweb.com)



Filters we applied

DNS– we observed DNS query & response packets

DNS- we observed DNS query & response packets

TCP – handshakes SYN, SYN-ACK,ACK

3/26/2026

[illegible]

Detecting suspicious connection attempts `tcp.flags.syn==1 && tcp.flags.ack==0` Retransmission - `tcp.analysis.retransmission`

[illegible]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Telnet

No.	Time	Source	Destination	Protocol	Length	Info
5.0	0.15451718	172.64.148.235	192.168.1.188	TLV1.2	80	Application Data
6.0	0.154778052	192.168.1.188	172.64.148.235	TLV1.2	84	Application Data
7.0	0.20646427	172.64.148.235	192.168.1.188	TCP	60	443 -> 58014 [ACK] Seq=25 Ack=29 Win=1 Len=0 Tval=1235883338 Tscir=75313825
72.1	0.203183186	192.168.1.188	172.64.148.235	TLV1.2	154	Application Data
72.2	0.204073829	172.64.148.235	192.168.1.188	TCP	60	443 -> 58014 [ACK] Seq=25 Ack=17 Win=10 Len=0 Tval=1235897973 Tscir=753128437
72.3	0.203189625	192.168.1.188	172.64.148.235	TLV1.2	154	Application Data
74.1	0.171509095	172.64.148.235	192.168.1.188	TCP	60	443 -> 58014 [ACK] Seq=25 Ack=20 Win=10 Len=0 Tval=1235898883 Tscir=753129567
78.1	0.204841959	172.64.148.235	192.168.1.188	TCP	140	Application Data
81.0	0.204841959	172.64.148.235	192.168.1.188	TCP	140	Application Data
81.1	0.204841959	172.64.148.235	192.168.1.188	TLV1.2	140	Application Data
81.2	0.204841959	172.64.148.235	192.168.1.188	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.1	0.162424762	172.64.148.235	192.168.1.188	TLV1.2	140	Application Data
83.2	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.3	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.4	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.5	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.6	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.7	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.8	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.9	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.10	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.11	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.12	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.13	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.14	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.15	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.16	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.17	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.18	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.19	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.20	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.21	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.22	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.23	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.24	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.25	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.26	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.27	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.28	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.29	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.30	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.31	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.32	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.33	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.34	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.35	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.36	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.37	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.38	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.39	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.40	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.41	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.42	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.43	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.44	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.45	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.46	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.47	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.48	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.49	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.50	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.51	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.52	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.53	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.54	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.55	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.56	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.57	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.58	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.59	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.60	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.61	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.62	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.63	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.64	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.65	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.66	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.67	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.68	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.69	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.70	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.71	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK] Seq=2058 Ack=254 Win=0 Len=0 Tval=123225060 Tscir=753099356
83.72	0.205093135	192.168.1.188	172.64.148.235	TCP	60	58014 -> 443 [ACK

Retransmission - tcp.analysis.retransmission

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcpdump -i eth0 -s 1500 -w -

Packet list Display filter Enter a display filter... Find Cancel

Options: [Barrow & Wide] Case sensitive Backwards Multiple occurrences

No.	Time	Source	Destination	Protocol	Length	Info
156	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
157	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
158	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
159	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
160	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
161	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
162	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
163	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
164	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
165	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
166	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
167	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
168	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
169	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
170	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
171	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
172	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
173	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
174	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
175	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
176	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
177	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
178	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
179	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
180	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
181	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
182	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
183	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
184	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
185	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=81981500 → 80 [RST] Seq=16460 Win=0 Len=0
186	24.184314424	192.168.1.1500	44.238.29.244	TCP	74	TCPPort=8198

CONCLUSION

6

In this project, Wireshark proved to be a powerful tool for understanding how networks operate. By capturing live traffic, it was possible to observe how protocols like TCP establish connections, how DNS resolves domain names, and how data is transmitted across the network. The use of filters made it easier to focus on specific types of traffic and identify important patterns.

Additionally, this practical demonstrated how Wireshark can be used not only for learning but also for troubleshooting and cybersecurity. It helped reveal how network issues such as retransmissions occur and how suspicious activities can be detected through unusual packet behavior.

Overall, the project showed that Wireshark is an essential tool for analyzing network communication, improving performance, and enhancing security. It provides deep visibility into network activity, making it valuable for students, network engineers, and cybersecurity professionals.

/26/2026